

Transient Filmwise Condensation and Interfacial Vapor Shear Effects on a Vertical Plate: Numerical Simulation and Parametric Analysis

Course: ME 647 – Vapor–Liquid Phase Change Phenomena.

Course instructor: Prof. Soumyadip Sett.

Group Members: Shrishti Mishra, Hari Balaji, and Diya Mehta

Abstract—Abstract— This study numerically analyzes filmwise condensation on a vertical plate under transient (stagnant vapor) and steady-state (with vapor shear) conditions.

Transient results, obtained using an FTBS scheme, closely match the Nusselt solution (0.64% mean error), with deviations near the leading edge. Increased subcooling thickens the film but reduces heat transfer.

In steady state, vapor shear thins the film and enhances heat transfer; however, gravity dominates up to 30 m/s, with a crossover velocity of 69 m/s.

I. INTRODUCTION

Filmwise condensation on cooled surfaces is a dominant heat transfer mode in power plant condensers, refrigeration systems, and process-industry heat exchangers. The classical steady-state theory of Nusselt [3] provides a closed-form solution under assumptions of laminar flow, pure conduction across the film, negligible inertia, and zero interfacial shear. However, practical situations often involve transient start-up behavior and non-negligible vapor velocities that impose a shear stress at the liquid–vapor interface.

The objectives of this work are twofold:

- 1) **Part A:** Develop and validate a transient numerical model for gravity-driven filmwise condensation and conduct parametric studies on the effect of wall subcooling ΔT and plate length L .
- 2) **Part B:** Extend the model to steady-state condensation with downward flowing vapor and quantify the effect of free-stream vapor velocity U_∞ on film thickness, local Nusselt number, and interfacial shear stress.

II. PROBLEM FORMULATION

A. Physical Configuration

A vertical flat plate of length L and width W is initially at saturation temperature T_{sat} . At $t = 0$ the plate temperature is instantaneously lowered to the uniform wall temperature $T_w < T_{\text{sat}}$, initiating condensation. A thin liquid film of local thickness $\delta(z, t)$ forms at the leading edge ($z = 0$) and drains downward under gravity. In Part B, a saturated vapor stream flows downward at free-stream velocity U_∞ , exerting an interfacial shear stress τ_i on the film surface.

B. Fluid Properties

All liquid properties are evaluated at the film temperature $T_f = (T_{\text{sat}} + T_w)/2 \approx 50^\circ\text{C}$; vapor properties are evaluated at saturation conditions ($\approx 60^\circ\text{C}$). The values used throughout are listed in Table I.

TABLE I: Fluid Properties (Water)

Property	Symbol	Value
Liquid density	ρ_l	988.0 kg/m^3
Liquid dyn. viscosity	μ_l	$5.47 \times 10^{-4} \text{ Pa}\cdot\text{s}$
Liquid kin. viscosity	ν_l	$5.54 \times 10^{-7} \text{ m}^2/\text{s}$
Thermal conductivity	k_l	$0.644 \text{ W}/(\text{m}\cdot\text{K})$
Specific heat	$c_{p,l}$	$4181 \text{ J}/(\text{kg}\cdot\text{K})$
Latent heat	h_{fg}	2382 kJ/kg
Vapor density	ρ_v	0.130 kg/m^3
Vapor dyn. viscosity	μ_v	$1.20 \times 10^{-5} \text{ Pa}\cdot\text{s}$
Vapor kin. viscosity	ν_v	$9.23 \times 10^{-5} \text{ m}^2/\text{s}$

III. PART A: TRANSIENT GRAVITY-DRIVEN CONDENSATION

A. Governing PDE Derivation

1) *Velocity Profile:* Under the Nusselt assumptions (laminar flow, negligible inertia, hydrostatic pressure), the momentum equation in the film reduces to:

$$\mu_l \frac{\partial^2 u}{\partial y^2} = -\rho_l g, \quad (1)$$

with boundary conditions $u|_{y=0} = 0$ (no-slip) and $(\partial u / \partial y)|_{y=\delta} = 0$ (zero shear at interface). This gives the parabolic velocity profile:

$$u(y) = \frac{\rho_l g}{\mu_l} \left(\delta y - \frac{y^2}{2} \right). \quad (2)$$

2) *Mass Flux and Continuity:* Integrating (2) across the film yields the mass flow rate per unit width:

$$\Gamma = \rho_l \int_0^\delta u dy = \frac{\rho_l^2 g \delta^3}{3\mu_l}, \quad \frac{d\Gamma}{dz} = \frac{\rho_l g}{\nu_l} \delta^2 \frac{\partial \delta}{\partial z}. \quad (3)$$

3) *Interface Energy Balance:* The condensation mass flux equals the conductive heat flux through the film [1]:

$$\dot{m}_{\text{cond}} = \frac{k_l \Delta T}{h_{fg} \delta} = \frac{\rho_l \alpha}{\delta}, \quad \alpha \equiv \frac{k_l \Delta T}{\rho_l h_{fg}}. \quad (4)$$

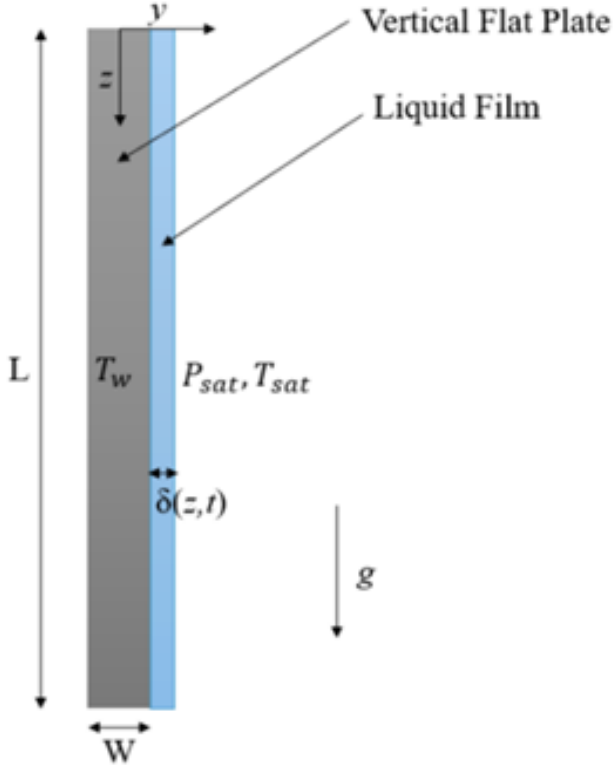


Fig. 1: Transient Gravity-Driven Filmwise Condensation

4) *Final PDE*: Combining continuity, (4), and the unsteady accumulation term gives the governing nonlinear hyperbolic PDE:

$$\frac{\partial \delta}{\partial t} + \underbrace{\frac{g \delta^2}{\nu_l}}_{C(\delta)} \frac{\partial \delta}{\partial z} = \underbrace{\frac{\alpha}{\delta}}_{S(\delta)} \quad (5)$$

where $C(\delta)$ is the kinematic wave speed and $S(\delta)$ is the condensation source term. The PDE is stiff near $\delta \rightarrow 0$ because $S \rightarrow \infty$.

B. Analytical Nusselt Baseline

Setting $\partial \delta / \partial t = 0$ in (5) and integrating gives the classical steady-state profile:

$$\delta_{\text{Nu}}(z) = \left(\frac{4 \nu_l \alpha z}{g} \right)^{1/4}, \quad (6)$$

with associated local and mean heat transfer coefficients:

$$h(z) = \frac{k_l}{\delta(z)}, \quad \bar{h} = \frac{4}{3} h(L). \quad (7)$$

C. Numerical Finite-Difference Scheme

1) *Spatial Discretization (FTBS Upwind)*: Since $C(\delta) \geq 0$ always, information propagates in the $+z$ direction. An upwind (backward-difference) scheme is used:

$$\frac{\partial \delta}{\partial z} \bigg|_i^n \approx \frac{\delta_i^n - \delta_{i-1}^n}{\Delta z}. \quad (8)$$

2) *Temporal Discretization: Hybrid Explicit–Implicit*: The convection term is treated explicitly (Forward Euler) to form an intermediate residual:

$$R_i = \delta_i^n - \Delta t C_i^n \frac{\delta_i^n - \delta_{i-1}^n}{\Delta z}, \quad (9)$$

while the source term is treated implicitly (Backward Euler) to prevent δ from becoming non-positive:

$$\delta_i^{n+1} - R_i = \frac{\Delta t \alpha}{\delta_i^{n+1}}. \quad (10)$$

Rearranging yields a quadratic in δ_i^{n+1} :

$$(\delta_i^{n+1})^2 - R_i \delta_i^{n+1} - \Delta t \alpha = 0, \quad (11)$$

whose positive root is taken as the physical solution.

3) *Stability: CFL Condition*: The time step is constrained by the Courant–Friedrichs–Lewy (CFL) condition:

$$\text{CFL} = \frac{C \Delta t}{\Delta z} \leq 1. \quad (12)$$

A conservative value of $\text{CFL} = 0.45$ is used throughout.

4) *Initial and Boundary Conditions*:

$$\delta(z, 0) = \delta_{\min} \approx 0 \quad (\text{dry plate at } t = 0), \quad (13)$$

$$\delta(0, t) = 0 \quad (\text{leading edge}). \quad (14)$$

A small seed value $\delta_{\min}$ regularises the singularity at $z = 0$.

D. Simulation Results and Validation

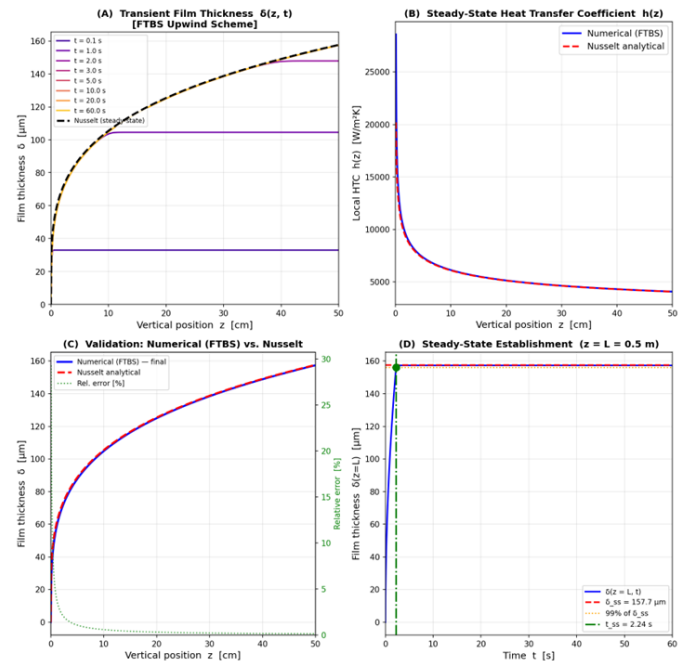


Fig. 2: Simulation Results and Validation at $L=0.5$ m and $\Delta T = 20\text{K}$

The numerical solution converges to within 0.64% of the Nusselt baseline across the plate. The 29.3% peak error at $z \approx 0$ is a singularity artifact and not physically significant. Key observations are:

TABLE II: Part A Validation Summary ($L = 0.5$ m, $\Delta T = 20$ K)

Quantity	Nusselt	Numerical
$\delta(z = L)$ [μm]	157.65	157.36
$\bar{h}$ [$\text{W}/(\text{m}^2 \text{ K})$]	5422.5	5489.4
Max. relative error	29.29 % (at $z \approx 0$)	
Mean relative error	0.64 %	
Steady-state time $t_{ss}(L)$ [s]	2.236	

- 1) The film grows non-uniformly. A kinematic wave propagates from $z = 0$; only the region behind the wave front reaches steady state.
- 2) $h(z) \propto z^{-1/4}$: the local HTC is highest at the leading edge and decreases monotonically.
- 3) Steady state is established within ≈ 2.2 s, confirming that the quasi-Nusselt approximation becomes valid almost immediately.

E. Parametric Study

1) *Effect of Wall Subcooling ΔT* : Table III summarizes results for $L = 0.5$ m and three subcoolings. Increasing ΔT raises α , causing more condensate to accumulate per unit area and thickening the film. Paradoxically, the mean HTC *decreases* with ΔT : the additional thermal resistance of the thicker film more than offsets the larger temperature driving force. The kinematic wave travels faster at higher α , reducing t_{ss} .

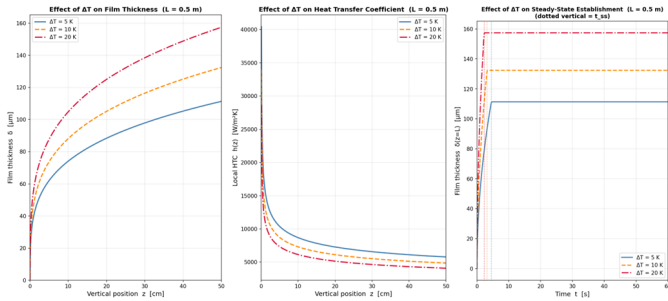


Fig. 3: Effect of Wall Subcooling ΔT ($L = 0.5$ m)

TABLE III: Effect of Wall Subcooling ($L = 0.5$ m)

ΔT [K]	$\delta(L)$ [μm]	$\bar{h}$ [$\text{W}/(\text{m}^2 \text{ K})$]	t_{ss} [s]
5	111.27	7763.2	4.473
10	132.32	6528.0	3.163
20	157.36	5489.4	2.236

2) *Effect of Plate Length L* : Table IV shows results for $\Delta T = 20$ K. The $\delta(z)$ profile is universal — all cases follow the same $z^{1/4}$ curve. A longer plate merely samples more of this curve, yielding a thicker terminal film and a lower spatially averaged HTC. The steady-state time scales as $t_{ss} \propto \sqrt{L}$, consistent with the kinematic wave travel time.

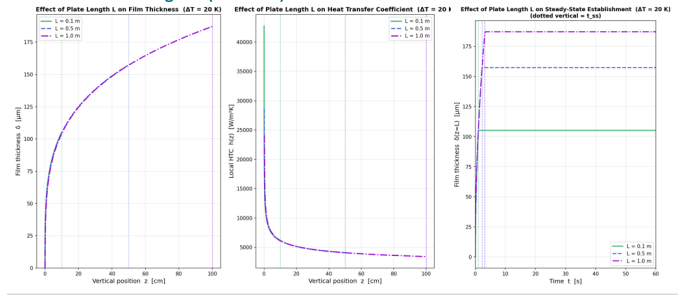


Fig. 4: Effect of plate length L ($\Delta T = 20$ K)

TABLE IV: Effect of Plate Length ($\Delta T = 20$ K)

L [m]	$\delta(L)$ [μm]	$\bar{h}$ [$\text{W}/(\text{m}^2 \text{ K})$]	t_{ss} [s]
0.1	105.24	8208.6	1.000
0.5	157.36	5489.4	2.236
1.0	187.13	4616.0	3.163

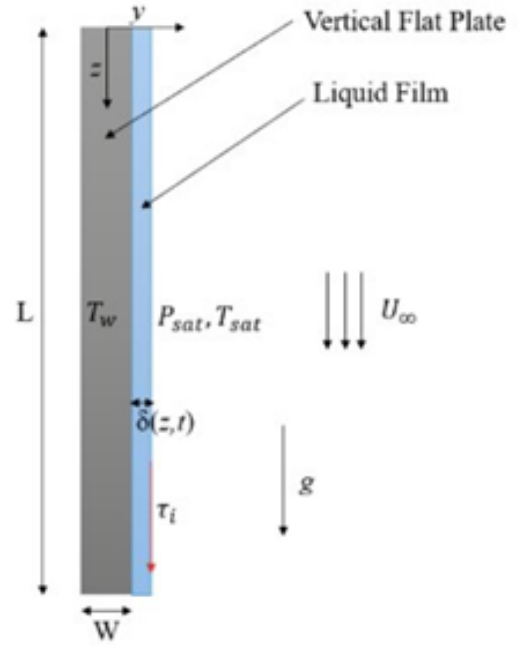


Fig. 5: Steady-State Condensation with Interfacial Vapor Shear

IV. PART B: STEADY-STATE CONDENSATION WITH INTERFACIAL VAPOR SHEAR

A. Modified Governing ODE

In Part B the vapor flows downward at constant free-stream velocity U_∞ , imposing an interfacial shear stress τ_i that acts as a second drainage mechanism in addition to gravity.

1) *Modified Velocity Profile*: The interfacial shear boundary condition replaces the zero-shear condition:

$$\mu_l \left. \frac{\partial u}{\partial y} \right|_{y=\delta} = \tau_i. \quad (15)$$

Solving the momentum equation gives:

$$u(y) = \frac{\rho_l g}{\mu_l} \left(\delta y - \frac{y^2}{2} \right) + \frac{\tau_i}{\mu_l} y, \quad (16)$$

and the modified mass flow rate per unit width:

$$\Gamma = \frac{\rho_l^2 g \delta^3}{3\mu_l} + \frac{\tau_i \rho_l \delta^2}{2\mu_l}. \quad (17)$$

2) *Blasius Interfacial Shear Correlation*: Assuming a laminar vapor boundary layer (Blasius), the local friction factor and shear stress are:

$$Re_{v,z} = \frac{\rho_v U_\infty z}{\mu_v}, \quad C_f = \frac{0.664}{\sqrt{Re_{v,z}}}, \quad \tau_i = \frac{1}{2} \rho_v C_f U_\infty^2. \quad (18)$$

3) *Final Steady-State ODE*: Combining the energy balance (4), the modified mass flux, and (18) gives:

$$\frac{d\delta}{dz} = \frac{k_l \mu_l (T_{\text{sat}} - T_w)}{\rho_l^2 g h_{fg} \delta^3 + \frac{1}{2} \rho_v C_f(z) U_\infty^2 \cdot \rho_l h_{fg} \delta^2} \quad (19)$$

or equivalently $d\Gamma/dz = \dot{m}_{\text{cond}}$, which can be written as:

$$\frac{d\delta}{dz} = \frac{k_l \mu_l \Delta T}{h_{fg} \delta^2 (\rho_l^2 g \delta + \rho_l \tau_i)}. \quad (20)$$

B. Numerical Solution

1) *Forward Euler Integration*: Since the problem is purely a steady-state spatial ODE, a simple first-order Forward Euler scheme suffices:

$$\delta_{i+1} = \delta_i + \Delta z \cdot f(z_i, \delta_i), \quad f(z_i, \delta_i) = \frac{k_l \Delta T \mu_l}{h_{fg} \delta_i^2 (\rho_l^2 g \delta_i + \rho_l \tau_i)}. \quad (21)$$

2) *Leading-Edge Regularisation*: At $z = 0$, $\delta = 0$ creates a division-by-zero singularity. The first grid point $z_1 = \Delta z$ is seeded with the pure-gravity Nusselt solution:

$$\delta_1 = \left(\frac{4 \nu_l \alpha z_1}{g} \right)^{1/4}. \quad (22)$$

3) *Grid Convergence*: A step size of $\Delta z = L/1000$ is used. Solutions at $N = 500$ and $N = 1000$ grid points agreed to within 0.5%, confirming convergence.

C. Results: Vapor Velocity Effects

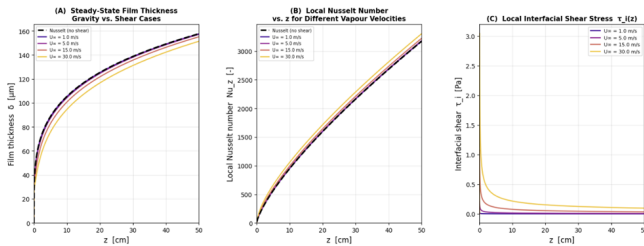


Fig. 6: Simulation Results

Table V presents the quantitative results for $U_\infty = 0$ –30 m/s.

TABLE V: Part B Results: Effect of Vapor Velocity ($L = 0.5$ m, $\Delta T = 20$ K)

U_∞ [m/s]	$Re_v(L)$ [-]	$\tau_i(L)$ [Pa]	$\delta(L)$ [μm]	$\Delta\delta$ vs. Nu	$\bar{h}$ [W/m ² K]
0 (Nu)	—	—	157.65	baseline	5430
1	5420	0.0006	157.60	−0.03 %	5432
5	27100	0.0066	157.12	−0.34 %	5489
15	81200	0.0341	155.15	−1.58 %	5685
30	162000	0.0964	151.40	−3.96 %	6020

D. Physical Interpretation

Film thinning. Higher U_∞ imposes greater interfacial shear, accelerating drainage. The film equilibrates at a smaller thickness where the increased drainage rate is balanced by condensation supply. The thinning effect is cumulative: a thinner film at the top remains thinner all the way to the bottom.

Heat transfer enhancement. Since $h = k_l/\delta$, any reduction in film thickness directly reduces the conduction resistance and increases the local HTC. At $U_\infty = 30$ m/s, the mean HTC increases from 5430 to 6020 W/(m² K), a gain of approximately 11%.

Why shear effects are small. Steam at 60 °C has a very low vapor density ($\rho_v = 0.130$ kg/m³). Even at $U_\infty = 30$ m/s, the dynamic pressure $\frac{1}{2} \rho_v U_\infty^2 \approx 0.059$ Pa is negligible compared with the gravitational body force $\rho_l g \delta \approx 1.52$ Pa. The film thickness deviation from the Nusselt baseline is only 3.96% at $U_\infty = 30$ m/s.

E. Critical Velocity U^*

The transition from gravity-dominated to shear-dominated drainage occurs when the plate-averaged shear-to-gravity ratio R equals unity:

$$R = \left\langle \frac{\tau_i(z)}{\rho_l g \delta(z)} \right\rangle = 1. \quad (23)$$

Scanning U_∞ from 0.01 to 100 m/s (500 log-spaced values) and solving (19) at each velocity yields:

$$U^* \approx 69.1 \text{ m/s}. \quad (24)$$

For $U_\infty \ll U^*$, the film is essentially at the Nusselt solution. For $U_\infty \gg U^*$, vapor shear substantially thins the film and significantly enhances heat transfer. The laminar assumption remains valid for all tested velocities since $Re_v(L) < 5 \times 10^5$.

V. DISCUSSION

A. Comparison of Part A and Part B Steady States

Both parts converge to the same Nusselt steady state when $U_\infty = 0$, confirming consistency of the two formulations. Part A solves the full transient PDE and reveals the time scale $t_{ss} \propto \sqrt{L}$ for the kinematic wave to traverse the plate, whereas Part B provides a rigorous steady-state spatial profile incorporating shear.

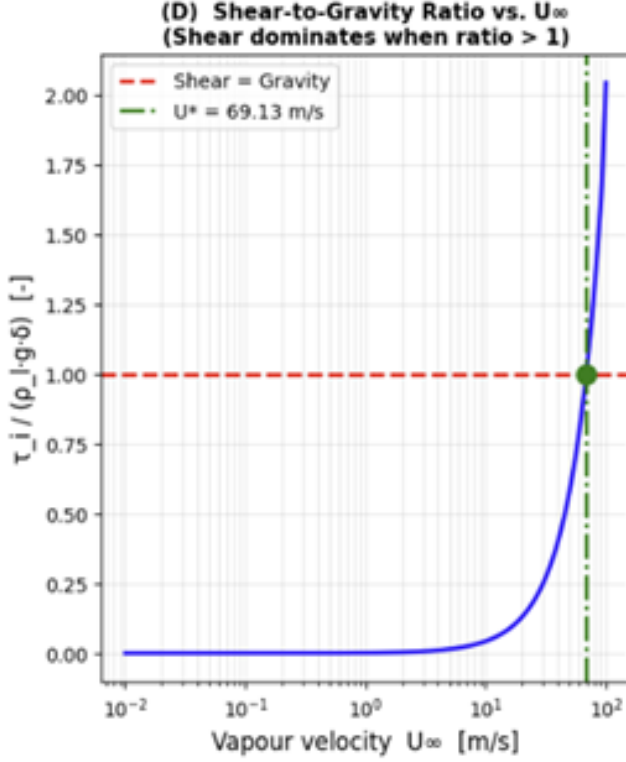


Fig. 7: Critical Velocity U^*

B. Limitations and Future Work

- 1) The Blasius correlation for τ_i assumes a laminar, undisturbed vapor boundary layer and neglects interaction with the liquid film. More accurate models (e.g., two-fluid boundary layer formulations) could be employed.
- 2) Subcooled liquid convection and vapor superheat corrections to h_{fg} (Jakob number corrections) are neglected but may matter at large ΔT .
- 3) Turbulent transition in the liquid film is not modelled; the laminar assumption should be checked for longer plates or higher ΔT .
- 4) Marangoni effects due to temperature-induced surface tension gradients are neglected.

VI. CONCLUSION

A numerically robust, transient finite-difference simulation of filmwise condensation on a vertical plate has been developed and validated against the classical Nusselt theory with a mean error of 0.64%. Key findings are:

- 1) Higher wall subcooling thickens the film but reduces the mean HTC; steady state is achieved faster because the kinematic wave speed is proportional to $\alpha \propto \Delta T$.
- 2) Plate length affects only the spatial extent of the universal $z^{1/4}$ film profile; longer plates have lower mean HTCs and longer transient times ($t_{ss} \propto \sqrt{L}$).
- 3) Downward vapor flow exerts interfacial shear that thins the condensate film and enhances heat transfer. For low-density steam at 60°C, the effect is modest ($< 4\%$ film thinning) for $U_\infty \leq 30$ m/s; the critical velocity at which shear equals gravity is $U^* \approx 69.1$ m/s.

- 4) The hybrid implicit–explicit FTBS scheme with a CFL = 0.45 constraint provides a stable and accurate solution to the otherwise stiff hyperbolic condensation PDE.

NOMENCLATURE

α	Condensation coefficient = $k_l \Delta T / (\rho_l h_{fg})$ [m ² /s]
$C(\delta)$	Kinematic wave speed = $g \delta^2 / \nu_l$ [m/s]
C_f	Blasius friction factor [-]
δ	Film thickness [m]
g	Gravitational acceleration, 9.81 m/s ²
$h, \bar{h}$	Local / mean heat transfer coefficient [W/(m ² K)]
h_{fg}	Latent heat of vaporisation [J/kg]
k_l	Liquid thermal conductivity [W/(m K)]
L	Plate length [m]
$\dot{m}_{\text{cond}}$	Condensation mass flux [kg/(m ² s)]
μ_l, μ_v	Dynamic viscosities, liquid/vapor [Pa s]
ν_l, ν_v	Kinematic viscosities, liquid/vapor [m ² /s]
Nu_z	Local Nusselt number = z/δ [-]
R	Plate-averaged shear-to-gravity ratio [-]
$Re_{v,z}$	Local vapor Reynolds number = $\rho_v U_\infty z / \mu_v$ [-]
ρ_l, ρ_v	Densities, liquid/vapor [kg/m ³]
$S(\delta)$	Condensation source term = α/δ [m/s]
τ_i	Interfacial shear stress [Pa]
T_{sat}, T_w	Saturation / wall temperature [K]
t_{ss}	Steady-state establishment time [s]
U^*	Critical vapor velocity (shear = gravity) [m/s]
U_∞	Vapor free-stream velocity [m/s]
y, z	Cross-film / streamwise coordinate [m]
Γ	Film mass flow rate per unit width [kg/(m s)]
ΔT	Wall subcooling = $T_{\text{sat}} - T_w$ [K]

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